

IBM APPLIED DATA SCIENCE CAPSTONE

Falcon 9 First-stage Landing Prediction

From launch records to geographic insights
and classification models

Cristian / Technical project report / September 2026



Photo: Michael Seeley / NASA APOD

Executive summary: findings and model performance

The strongest conclusion is a testable workflow, with clear limits on generalization.

Finding	Evidence	Interpretation
Landing outcomes vary	60 / 90 Class 1 outcomes	Payload, orbit, site and time matter jointly.
Three models tie on test data	83.33% accuracy; 15 / 18 correct	LR, SVM and KNN outperform the 66.67% baseline.
KSC leads the dashboard sample	10 / 13 Class 1; 76.92%	This is a separate 56-record cohort.

Reported model scores are laboratory results: scaling before the split introduces leakage.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Can launch information predict a landing outcome?

Business context: first-stage recovery is relevant to launch economics and reuse.

- 01** Define a binary target from the recorded first-stage landing outcome.
- 02** Explore payload, orbit, launch site, booster history and time.
- 03** Compare interpretable baselines with tuned classification models.
- 04** Use the evidence to guide further validation before operational decisions.

Three snapshots answer different questions

Keep denominators and outcome definitions explicit when comparing results.

Analysis	Records / labels	Coverage
Python EDA + ML	90 launches • 3 site labels	4 Jun 2010–5 Nov 2020
SQL exercises	101 launches • 4 site labels	4 Jun 2010–6 Dec 2020
Folium + Dash	56 launches • 4 site labels	Separate course snapshot

Do not transfer a percentage from one snapshot to another.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

SpaceX API: build one record per eligible launch

The notebook uses a fixed API snapshot and enriches nested identifiers.

- 01** Load launch JSON; flatten the records with `pandas.json_normalize`.
- 02** Keep single-core, single-payload launches; parse the UTC date.
- 03** Resolve rocket, launchpad, payload and core IDs through helper calls.
- 04** Filter to Falcon 9; reset flight numbers and export `dataset_part_1.csv`.

Web scraping provides a second historical view

A fixed Wikipedia revision makes the intended source reproducible.

- 01** Fetch the Falcon 9 / Falcon Heavy launch-list HTML with requests.
- 02** Use BeautifulSoup to identify launch tables and clean their headers.
- 03** Parse date, booster, payload, customer, site and outcome fields.
- 04** Export spacex_web_scraped.csv for tabular inspection.

The target definition controls the interpretation

Mean-impute payload; retain meaningful missing landing-pad identifiers.

Recorded landing outcome	Records	Class
True ASDS / True RTLS / True Ocean	$41 + 14 + 5 = 60$	1
False or unknown / no attempt	30	0
LandingPad is missing	26 / 90	Retain
Payload mean used for imputation	6,104.96 kg	Numeric input

Five Class 1 records are ocean landings: the label is broader than recovered boosters.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Use each analytical tool for a distinct question

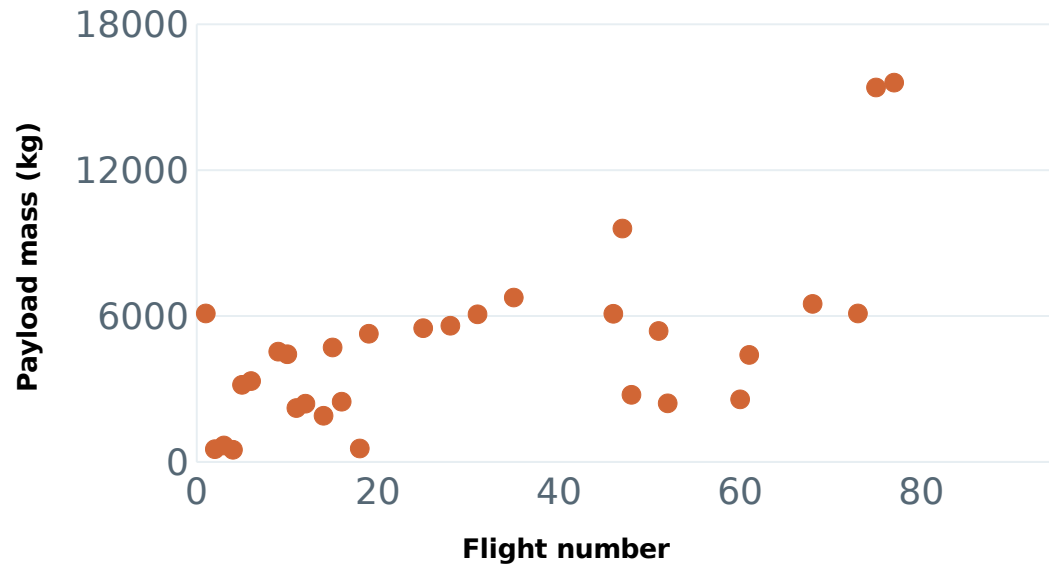
Combine quantitative comparisons with interactive geographic and mission views.

Tool	Method	Question
Python / pandas	Group by site, orbit, year; visualize	Where do Class 1 rates differ?
SQLite	Filter, aggregate, rank and compare	What do detailed launch records show?
Folium	Markers, clusters, distance lines	How are sites positioned geographically?
Plotly Dash	Site selector + payload range + callbacks	How do filtered outcomes change?

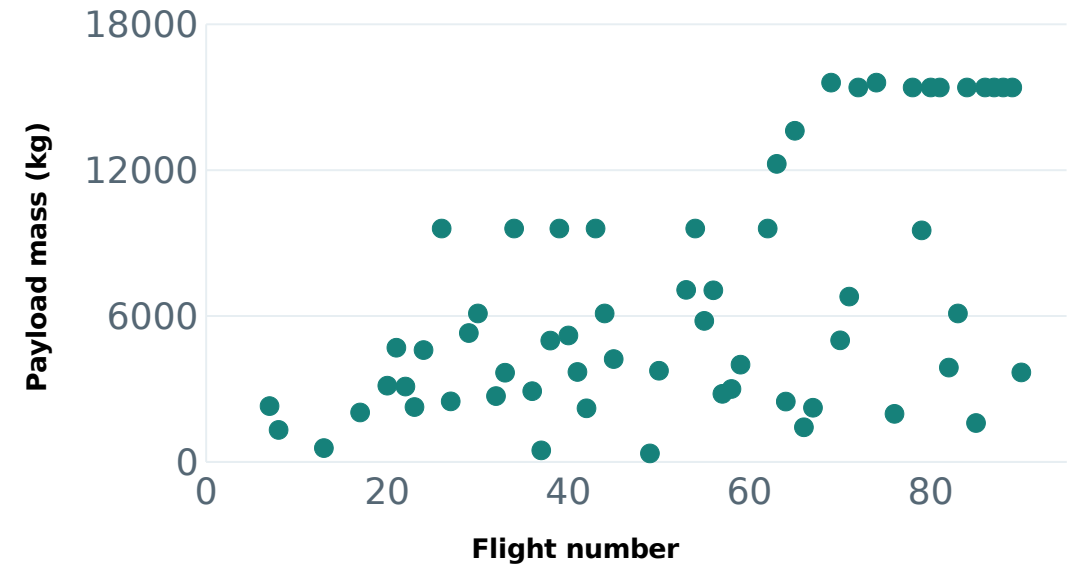
Payload alone does not separate the two classes

Later flights include many Class 1 outcomes across a wide payload range.

Class 0 (n=30)



Class 1 (n=60)



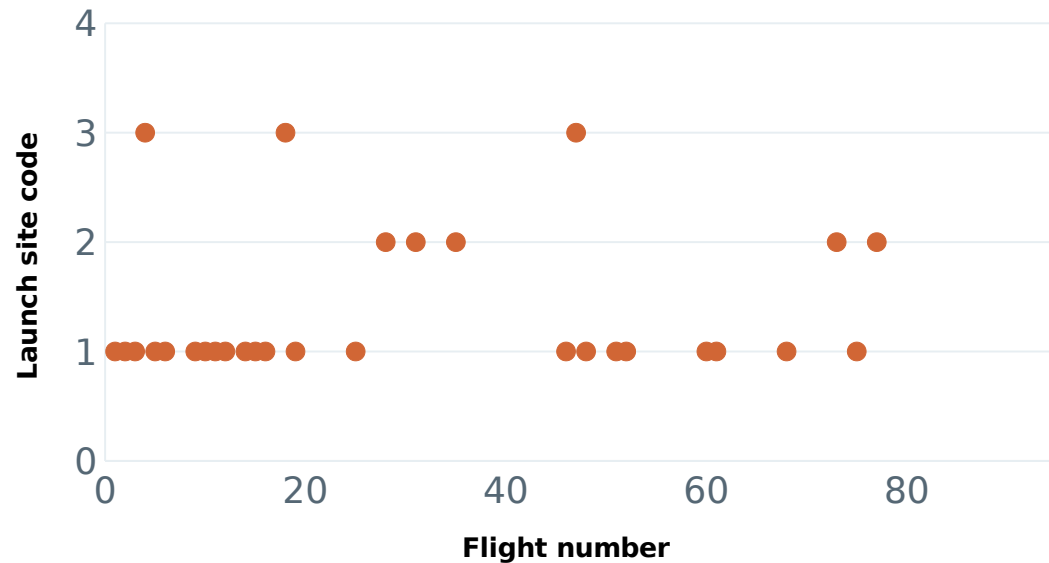
Flight number also represents time: apparent payload effects may reflect operational evolution.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

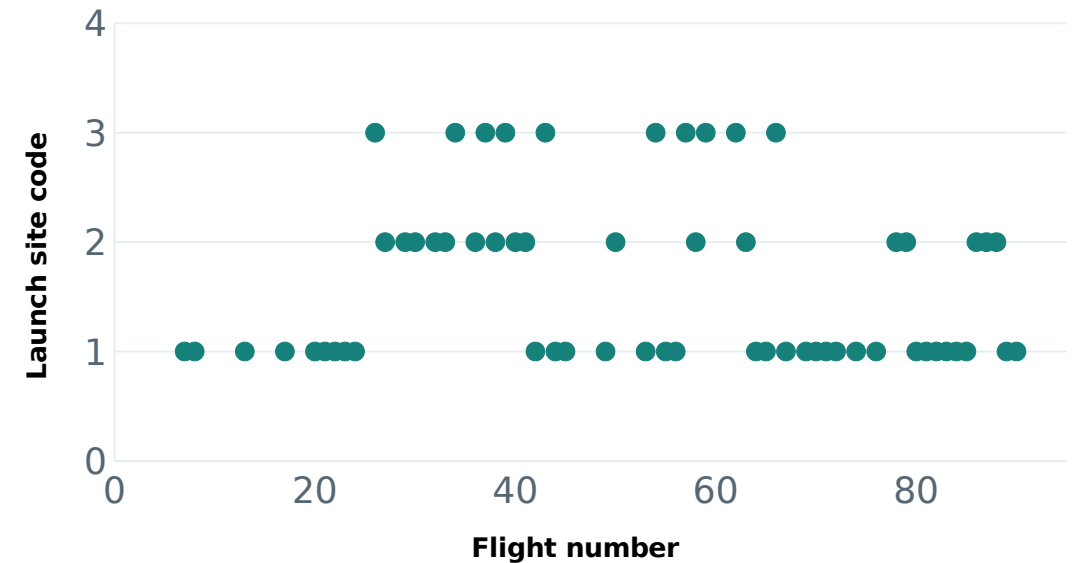
Flight number by launch site

Site codes: 1 = CCAFS SLC 40; 2 = KSC LC 39A; 3 = VAFB SLC 4E.

Class 0 (n=30)



Class 1 (n=60)



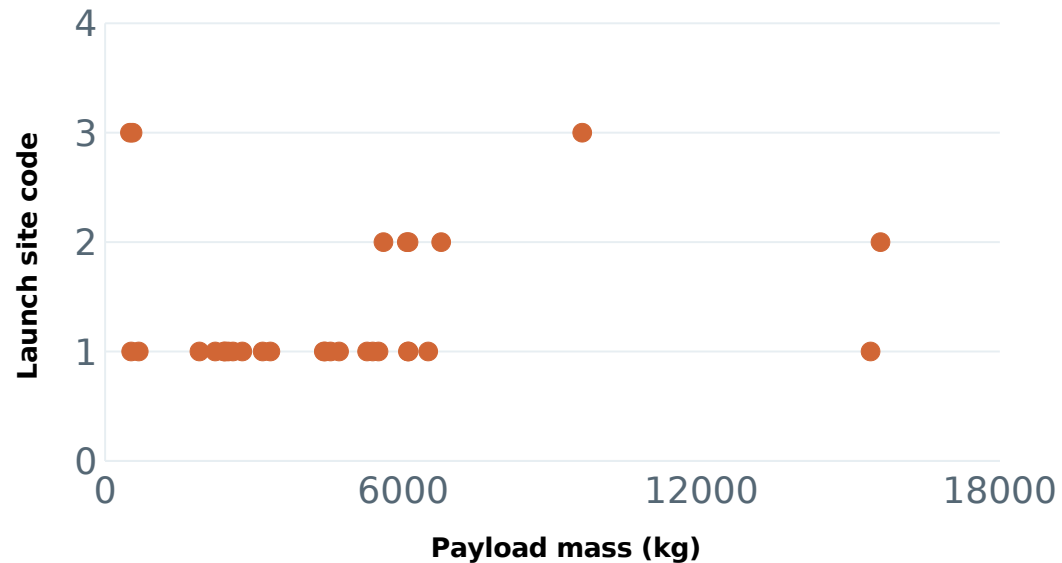
Sites serve different periods of the launch program. Compare site rates with time in view.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

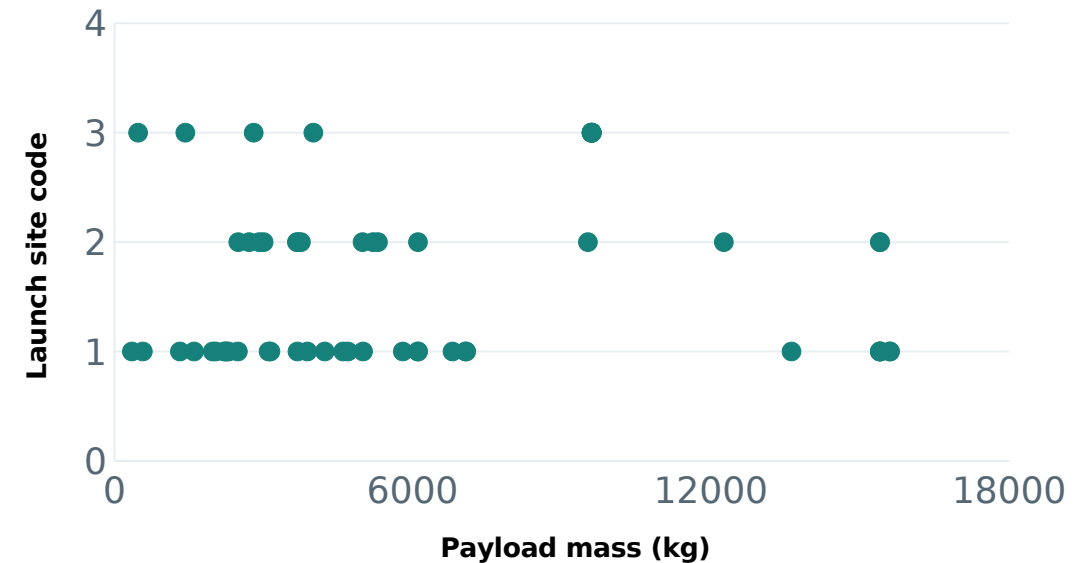
Payload mass (kg) by launch site

Site codes: 1 = CCAFS SLC 40; 2 = KSC LC 39A; 3 = VAFB SLC 4E.

Class 0 (n=30)



Class 1 (n=60)



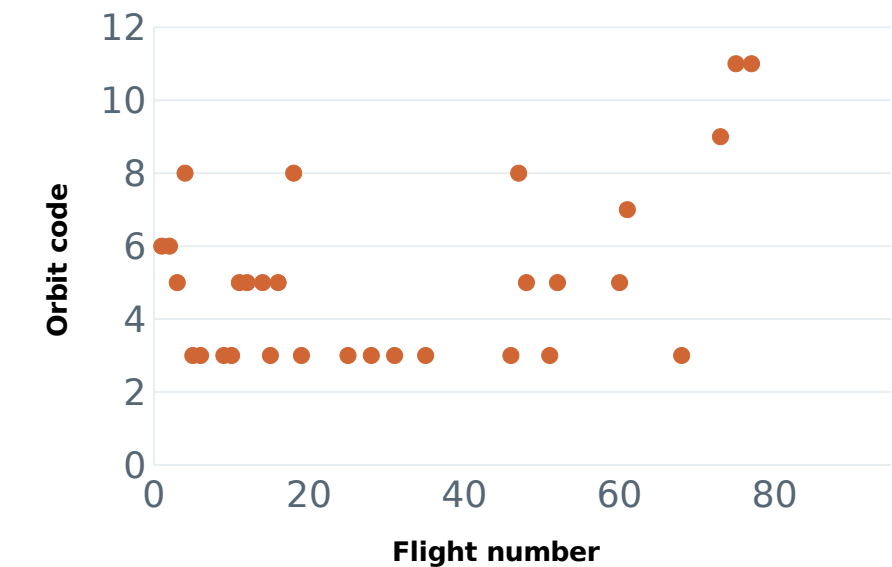
Both classes span multiple sites and payload ranges. Site alone does not separate outcomes.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

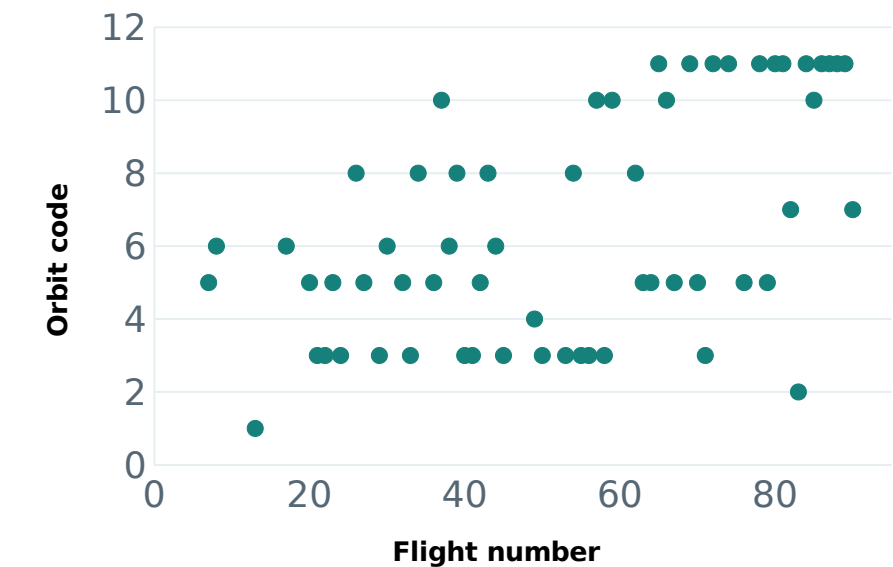
Flight number by orbit

Orbit codes identify categories, not ordered orbital distances.

Class 0



Class 1



Code	Orbit
1	ES-L1
2	GEO
3	GTO
4	HEO
5	ISS
6	LEO
7	MEO
8	PO
9	SO
10	SSO
11	VLEO

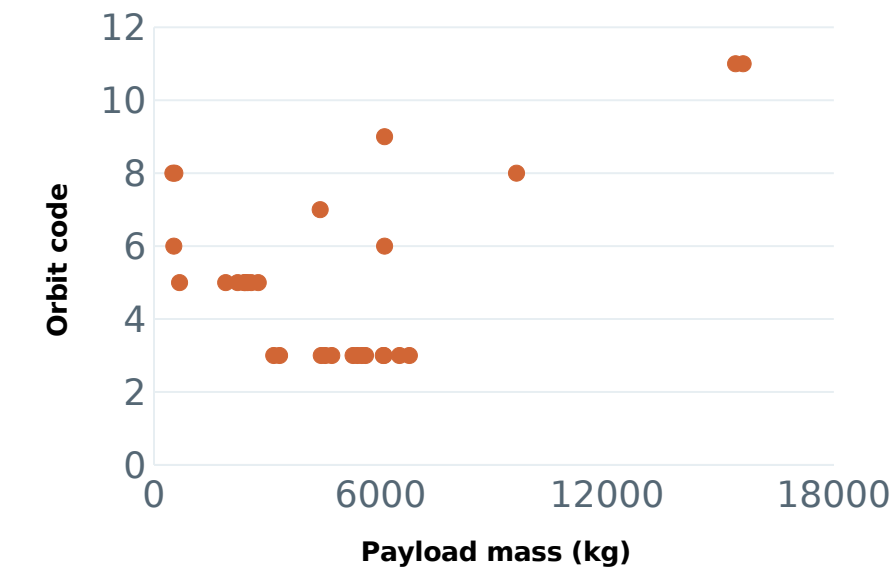
The orbit mix changes across flight numbers, so time can confound orbit comparisons.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

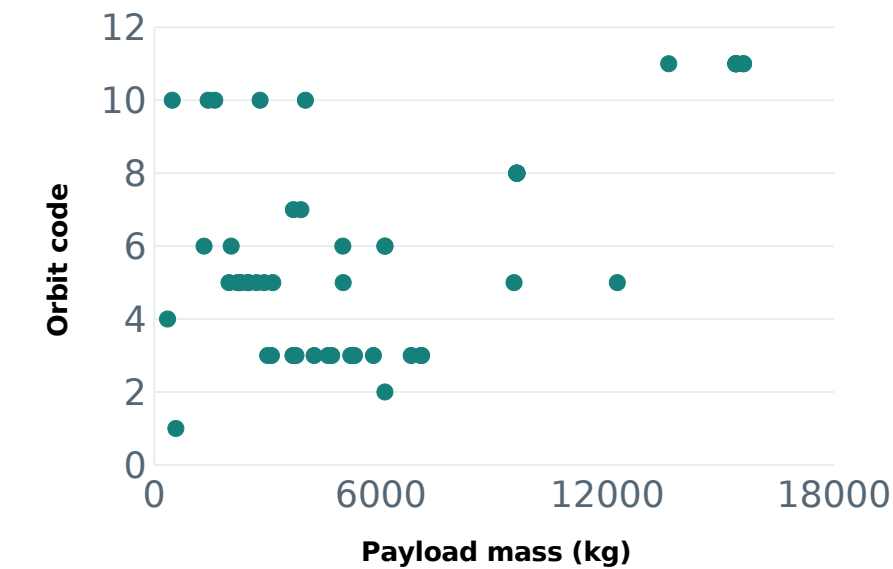
Payload mass (kg) by orbit

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Class 0



Class 1



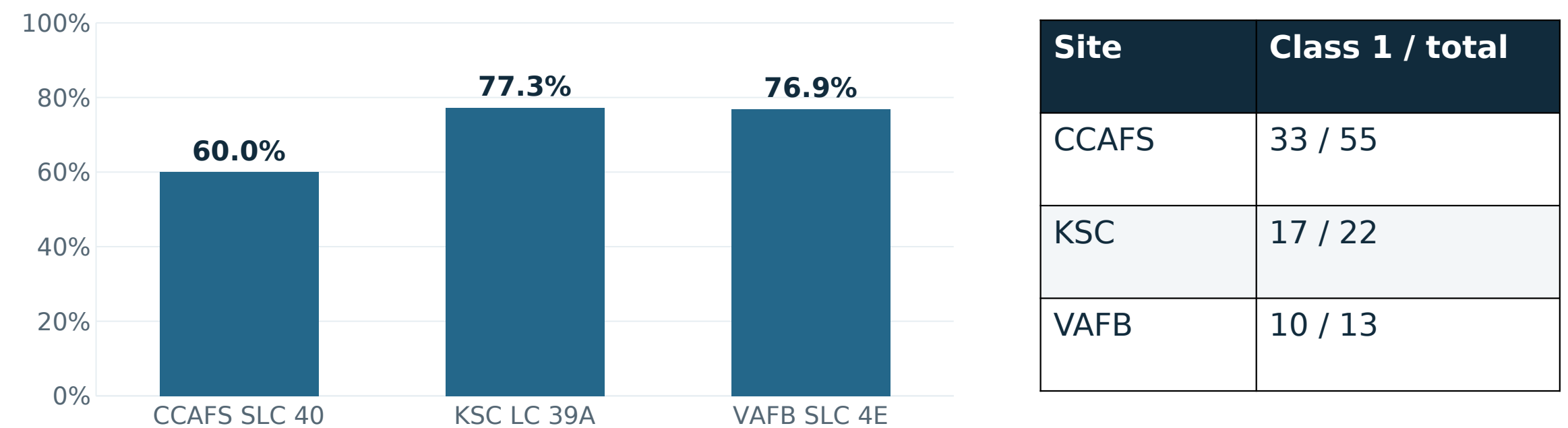
Code	Orbit
1	ES-L1
2	GEO
3	GTO
4	HEO
5	ISS
6	LEO
7	MEO
8	PO
9	SO
10	SSO
11	VLEO

Payload distributions depend on orbit. Marginal payload rates combine different mission types.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

The largest success count is not the highest rate

CCAFS has more Class 1 records; KSC and VAFB have higher observed rates.

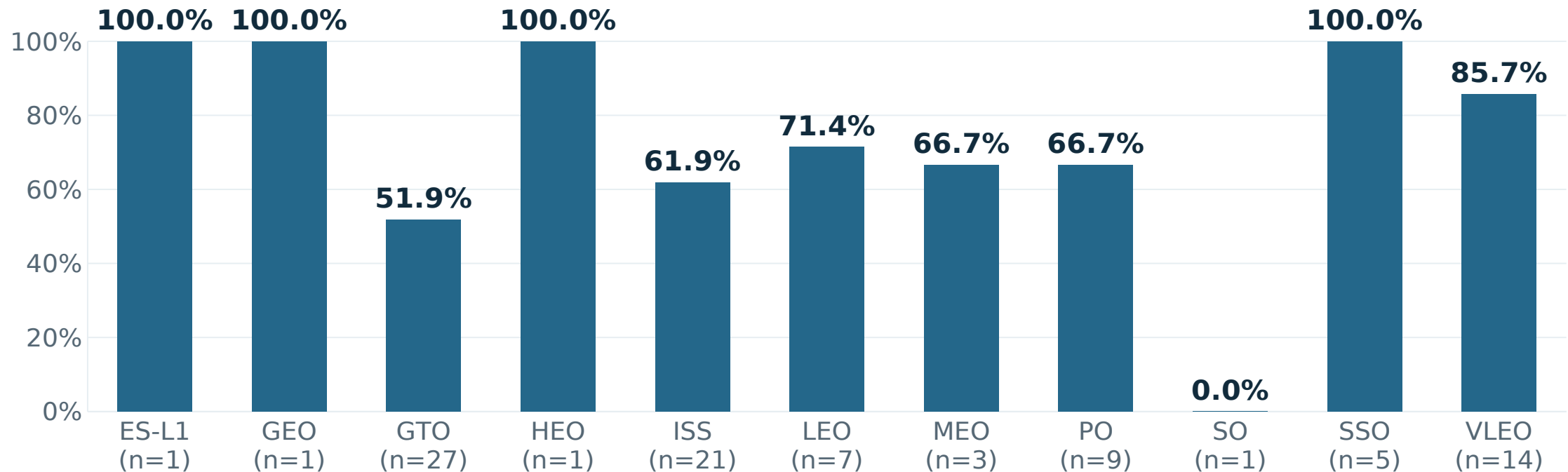


KSC (77.27%) and VAFB (76.92%) differ by only 0.35 percentage points.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Orbit rates need their sample sizes

SSO and VLEO show high observed rates; several 100% groups contain only one launch.

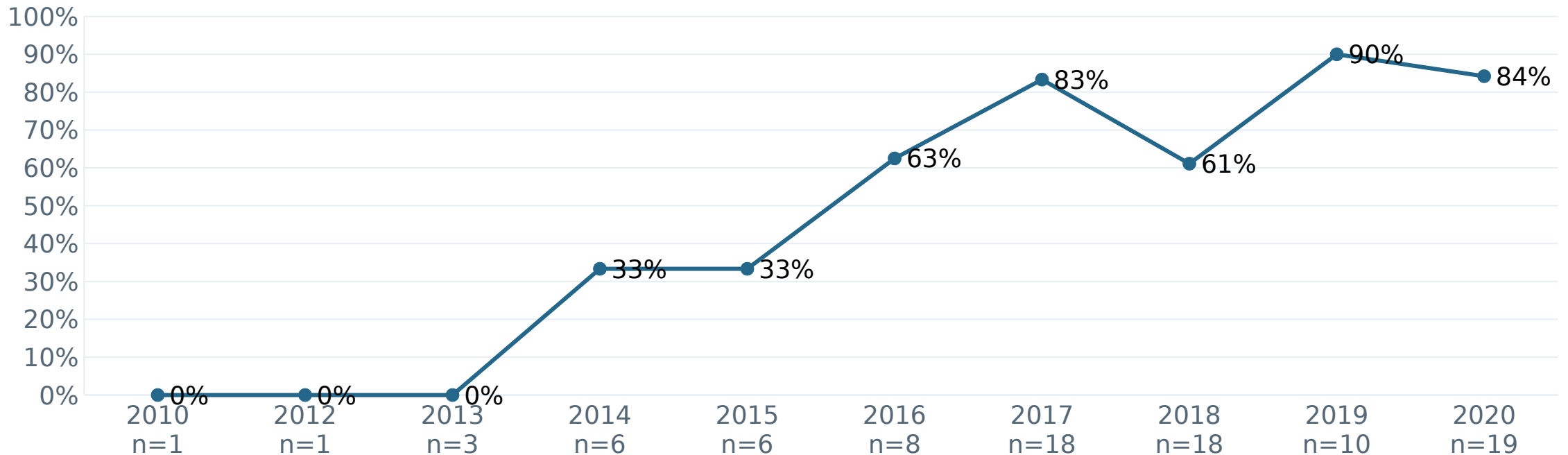


GTO: 14 / 27 = 51.85% • SSO: 5 / 5 • VLEO: 12 / 14 = 85.71%

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Performance improves over time, but not monotonically

The late-period gains coexist with a visible decrease in 2018.



2019: 9 / 10 = 90% • **2020 snapshot: 16 / 19 = 84.21%**

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

EDA with SQL: queries and definitions

SQLite supports auditable filters, aggregates and rankings.

Operation	Implementation	Purpose
Import and inspect	pandas.read_csv; to_sql; PRAGMA	101 launch records with 10 fields
Clean text	TRIM(Mission_Outcome)	Merge the trailing-space Success label
Aggregate	SUM, AVG, MIN, MAX, GROUP BY	Payload statistics and first landing date
Rank and filter	ORDER BY; WHERE; strftime	Site comparisons and historical events

SQL landing success here means `TRIM(Landing_Outcome) LIKE 'Success%'`.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

SQL turns launch records into auditable facts

Aggregation results from the 101-row SQL snapshot.

Question / operation	Verified result
SUM(payload) for NASA (CRS)	45,596 kg
AVG(payload) for F9 v1.1 variants	2,534.67 kg
MIN(date) for successful ground-pad landing	22 December 2015
MAX(payload) across the SQL snapshot	15,600 kg
COUNT of distinct launch-site labels	4

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Filtering reveals specific boosters and failure events

Successful drone-ship landings with $4,000 < \text{payload} < 6,000$ kg.

Query	Returned boosters
Landing_Outcome = Success (drone ship) AND payload > 4000 AND payload < 6000	F9 FT B1022 F9 FT B1026 F9 FT B1021.2 F9 FT B1031.2

2015 drone-ship failures

January: B1012 • April: B1015 • Both at CCAFS LC-40

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Source: supplied SQL notebook; queries rerun on IBM Spacex.csv (n=101).

Mission success and landing success are different

A payload can reach its mission objective even when first-stage landing fails.

Mission outcome after TRIM	Records
Success	99
Success (payload status unclear)	1
Failure (in flight)	1

A small cleaning issue matters

The raw SQL output separates “Success” and “Success ”.

Trimming merges 98 + 1 rows without changing the other outcome labels.

Do not substitute Mission_Outcome for the first-stage classification target.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

KSC leads the SQL landing-success rate

All 101 SQL records. Numerator: landing outcomes beginning with Success.

Launch site	Success / launches	Landing success rate
KSC LC-39A	20 / 25	80.00%
CCAFS SLC-40	25 / 34	73.53%
VAFB SLC-4E	10 / 16	62.50%
CCAFS LC-40	6 / 26	23.08%

CCAFS SLC-40 has the largest success count (25). KSC has the highest rate (80%).

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Historical outcome ranking and launch growth

The ranking uses the lab date window. Annual counts use the full SQL snapshot.

Landing outcome	Count
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

4 Jun 2010 to 20 Mar 2017

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Year	Launches	Success landings
2010	2	0
2012	2	0
2013	3	0
2014	6	0
2015	7	1
2016	8	5
2017	18	14
2018	20	10
2019	11	10
2020	24	21

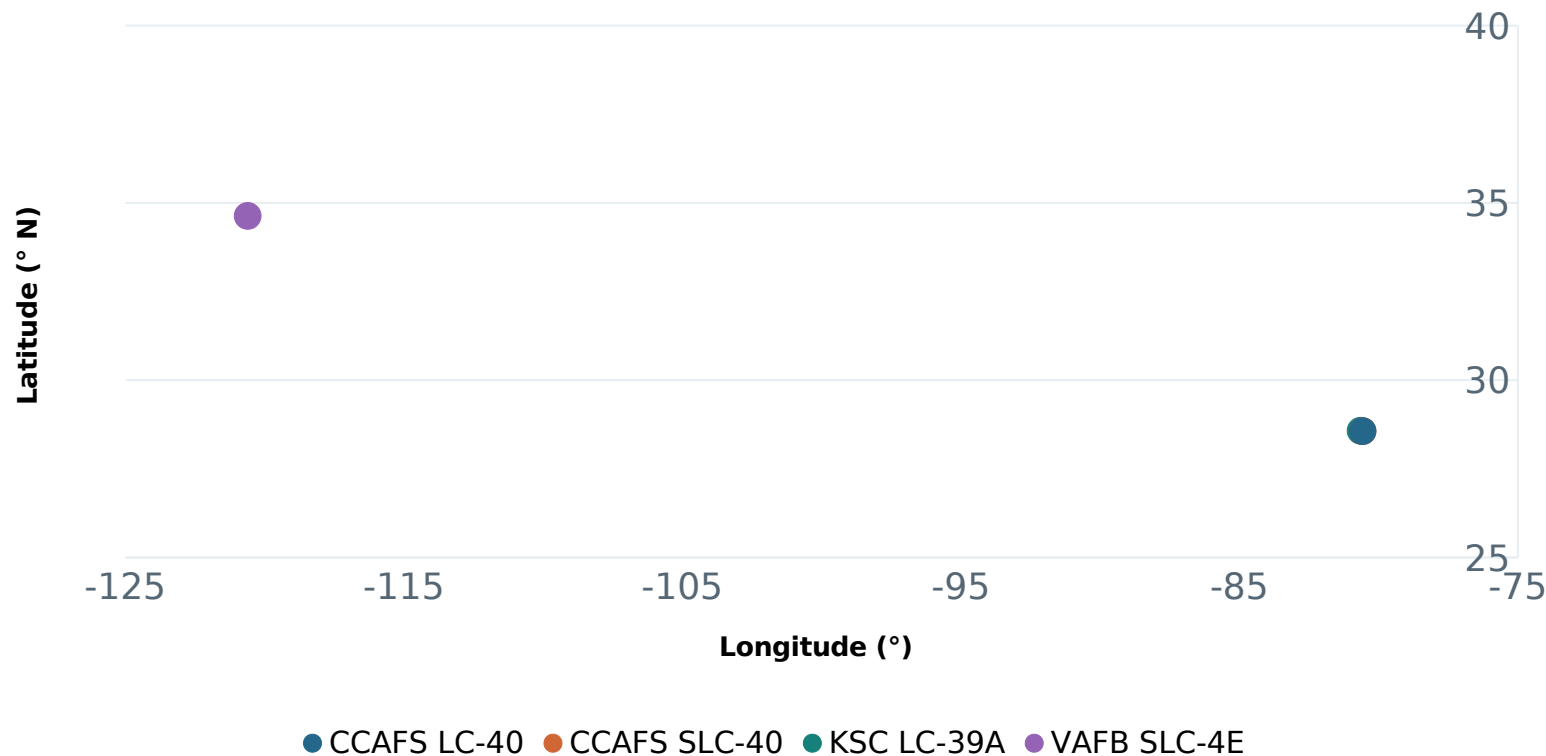
Folium and Dash: implementation and controls

Geography and launch filtering answer complementary questions.

Component	Implementation	Result
Folium locations	Map + CircleMarker + labels	Four source site labels
Folium outcomes	MarkerCluster + green/red Marker	56 records; green=1, red=0
Folium proximity	Haversine + PolyLine	Distances to four selected points
Dash dropdown + callback	site-dropdown → success-pie-chart	All-site counts or site outcome split
Dash range + callback	payload-slider + site-dropdown	Payload / outcome by booster

Folium inputs: launch sites on two coasts

Coordinate overview. The companion HTML contains the interactive basemap.



Geographic interpretation

All labels lie near a coast.

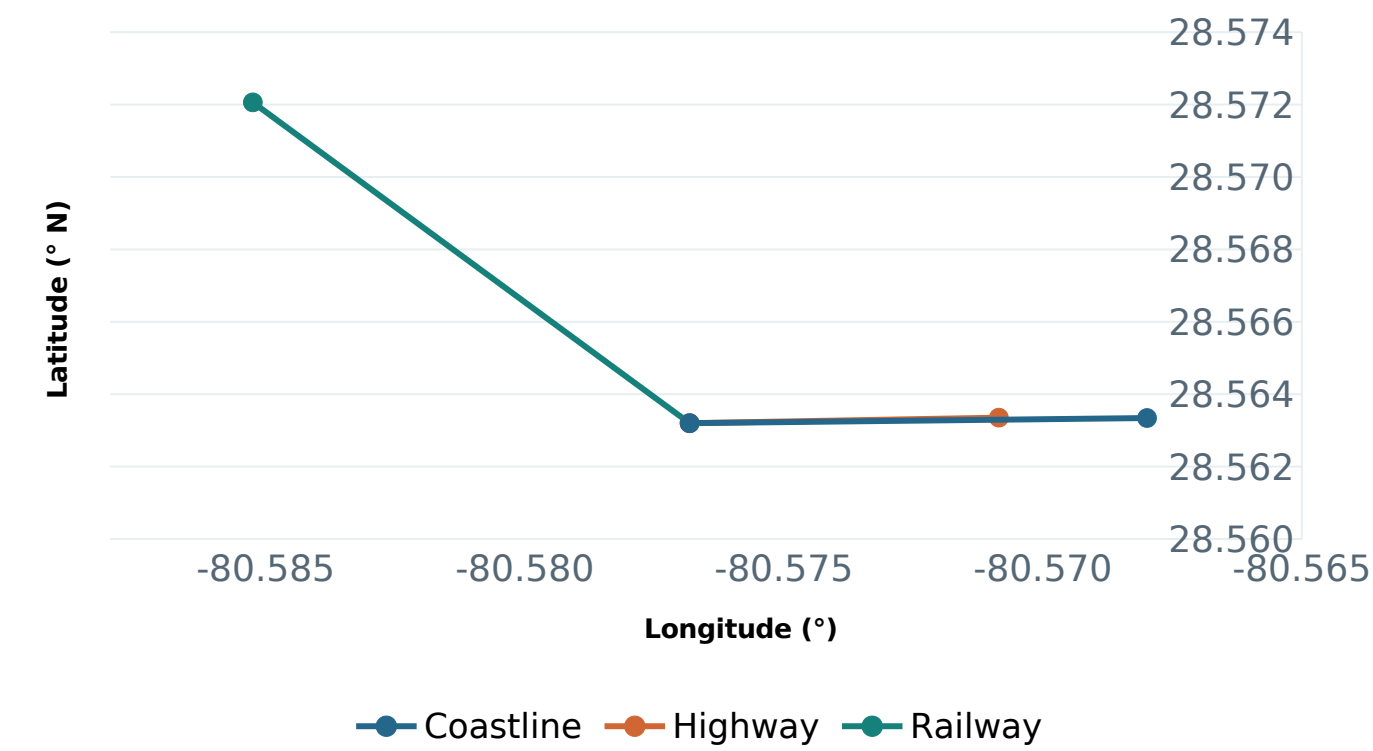
Latitudes are about 28–35° N: these sites are not on the Equator.

Ocean-facing trajectories are consistent with a safety rationale.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Folium proximity results: selected reference points

CCAFS SLC-40. Coordinate diagram below; interactive basemap in companion HTML.



Selected feature	Distance
Highway point	0.583 km
Coastline point	0.863 km
Railway point	1.285 km
City point	51.434 km

Manual points, not verified nearest features or driving routes.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Launch records and geographic interpretation

56 records use clustered markers. Identical coordinates expand on cluster selection.

Site label	Class 1	Class 0	Total
CCAFS LC-40	7	19	26
CCAFS SLC-40	3	4	7
KSC LC-39A	10	3	13
VAFB SLC-4E	4	6	10

Proximity interpretation

Selected road, rail and coastline points are within 1.3 km of the example origin.

The selected city point is 51.4 km away. This does not prove the nearest-city distance for all sites.

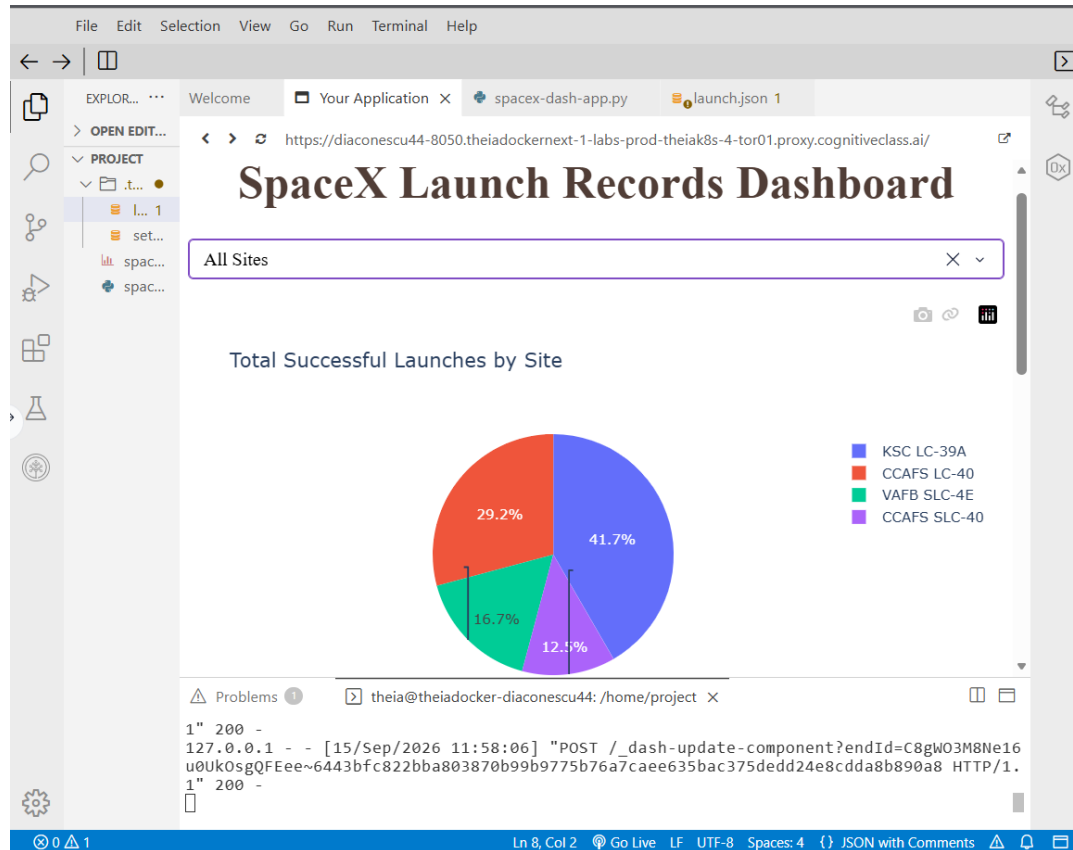
Interactive map: reports/SpaceX_Folium_Interactive.html

The PDF includes coordinate diagrams; a rendered basemap capture is still needed.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

KSC leads the 56-record dashboard sample

The all-sites pie shows the distribution of Class 1 counts.



Two different percentages

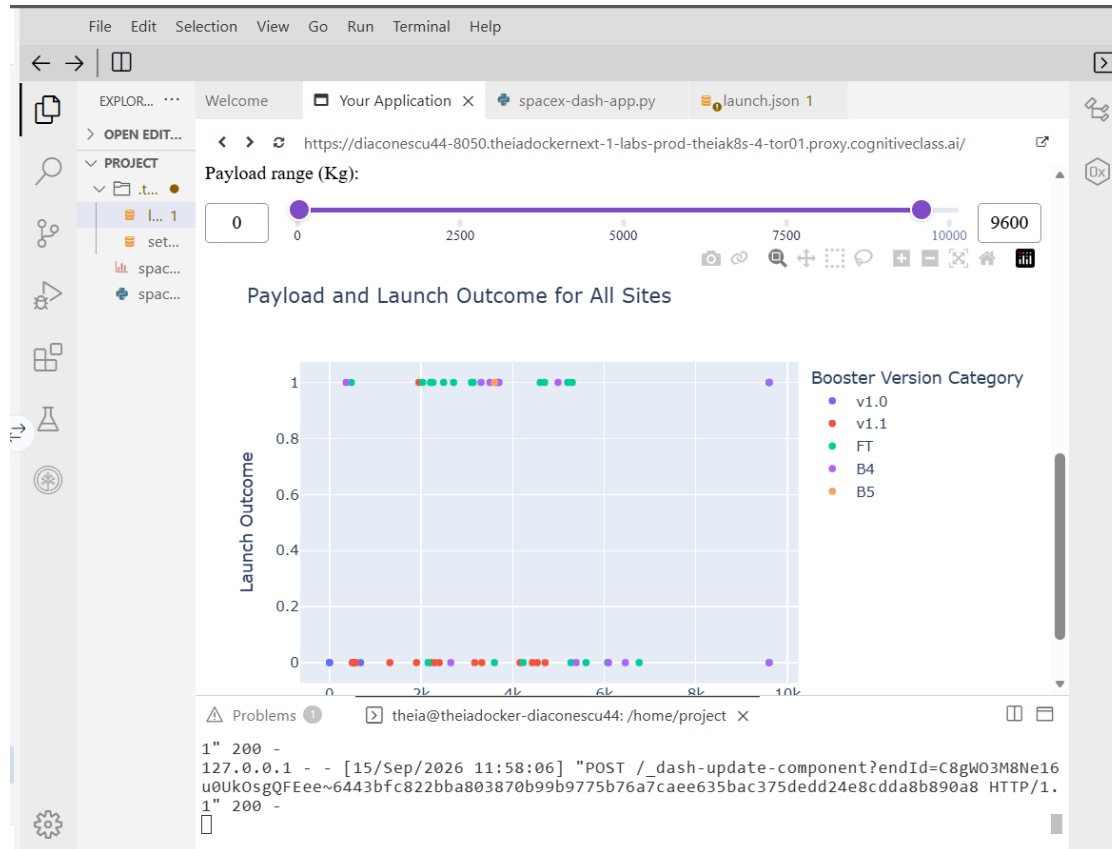
41.7% = 10 / 24
KSC share of all Class 1 records

76.9% = 10 / 13
Class 1 rate within KSC

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Payload filtering makes comparisons interactive

Site selection and a payload range control the scatter plot.

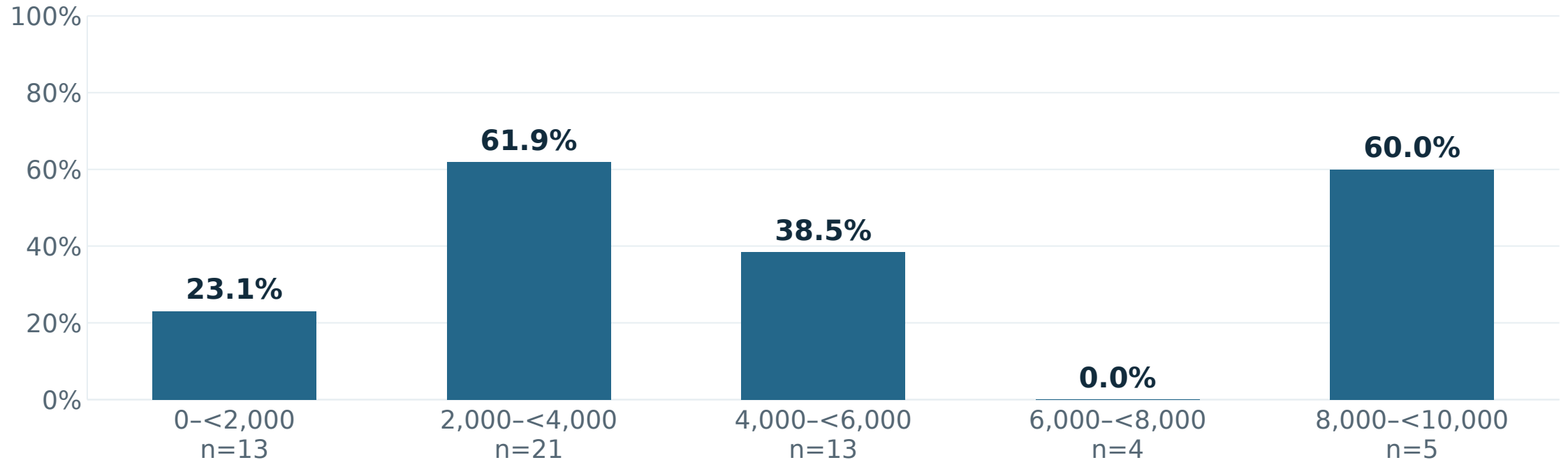


- 01** X: payload mass (kg).
Y: outcome class, 0 or 1.
- 02** Color: booster version category.
- 03** Filters help inspect subsets; they do not establish causation.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Payload-band rates are sensitive to the denominator

Recomputed fixed 2,000 kg bands for the dashboard's 56 records.



Highest: 2,000-<4,000 kg (13 / 21). Lowest: 6,000-<8,000 kg (0 / 4).

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

B5 has 100% from one observed launch

FT provides the largest body of positive outcomes in this dashboard snapshot.

Booster category	Class 1 / total	Observed rate
v1.0	0 / 5	0.00%
v1.1	1 / 15	6.67%
FT	16 / 24	66.67%
B4	6 / 11	54.55%
B5	1 / 1	100.00%

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Tune four classifiers on an 80/20 split

90 labels • 83 encoded input features • 72 train records • 18 test records

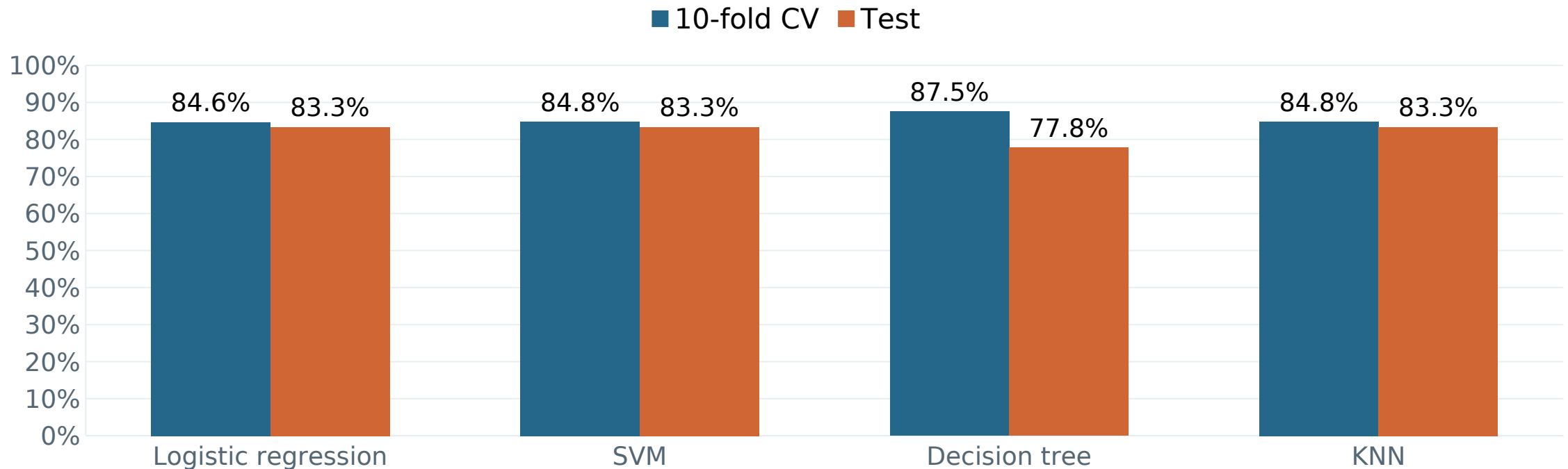
- 01 One-hot encode categorical inputs and convert the design matrix to float.
- 02 Standardize X; split with test_size=0.2 and random_state=2.
- 03 Use GridSearchCV with 10-fold cross-validation for each model.
- 04 Evaluate the selected estimator on test accuracy and a confusion matrix.

Lab limitation: the scaler is fitted before the split. Future work must fit it inside training folds.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Three models tie at 83.33% test accuracy

The tree wins cross-validation, but has the lowest saved test score.



Test baseline: 66.67% (always predict Class 1). One test record changes accuracy by 5.56 points.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Best hyperparameters from the saved grid searches

These settings describe the submitted laboratory run.

Model	Selected hyperparameters
Logistic regression	C=0.01; penalty=l2; solver=lbfgs
SVM	kernel=sigmoid; C=1; gamma $\approx$ 0.03162
Decision tree	criterion=gini; max_depth=6; max_features=sqrt min_samples_leaf=4; min_samples_split=10; splitter=random
KNN	n_neighbors=10; p=1 (Manhattan); algorithm=auto

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Errors are optimistic predictions of Class 1

Rows are actual classes; columns are predicted classes.

LR / SVM / KNN

Actual ↓ / predicted →	0	1
0	3	3
1	0	12

Class 1 precision: 80%

Class 1 recall: 100% • specificity: 50%

Decision tree

Actual ↓ / predicted →	0	1
0	2	4
1	0	12

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Best model: a tie on the holdout sample

LR, SVM and KNN each classify 15 of 18 test records correctly.

Selection criterion	Evidence	Decision
Holdout accuracy	LR / SVM / KNN: 83.33%	No unique winner on this split
Cross-validation score	Tree: 87.50%	CV leader, but test score is 77.78%
Error profile	Tied models: 3 false positives	All have specificity of only 50%
Practical starting model	Logistic regression	Simple reference model for a clean rerun

Prefer LR provisionally for simplicity. Confirm selection after leakage-free validation.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Before deployment, strengthen the evaluation

The next evaluation must remove leakage and test future-launch performance.

Observed limitation	Next validation step
Scaling before train/test split	Fit StandardScaler inside a cross-validated Pipeline.
Repeated booster serials; time trends	Add grouped and chronological holdouts.
18 test records; tied model scores	Report uncertainty and repeat validation on new launches.
Invalid tree grid option; no tree seed	Remove auto; set random_state; save versions and outputs.
Some fields may be updated after launch	Audit feature availability at prediction time.

Better definitions can improve decisions more than a leaderboard

Three distinctions prevent misleading conclusions.

- 01 Prediction target: separate landing, recovered hardware and reusable hardware before modeling economics.
- 02 Comparison unit: keep cohort, sample size and site naming consistent before ranking locations.
- 03 Decision risk: false positives dominate the saved errors; evaluate their cost before selecting a threshold.

The project connects collection, geography and prediction

A reproducible analytical story matters as much as the final score.

- 01** EDA reveals associations with orbit, site and operational history; none is established as causal.
- 02** Folium and Dash make geographic context and filtered outcomes easier to explore.
- 03** LR, SVM and KNN reach 83.33% in the saved test run; tree reaches 77.78%.
- 04** Next priority: remove leakage, validate across time and align the label with the intended decision.

Project repository and accompanying files

The code package preserves the supplied notebooks and adds runnable companion scripts.

GitHub repository

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Directory	Contents
notebooks/	Seven supplied project notebooks
app/ and sql/	Dashboard script; original SQL exercises; report queries
data/	Six course CSV snapshots with source information
reports/	Revised PDF / PowerPoint; interactive Folium HTML

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Submission requirements and evidence

The presentation separates methods, observed results and recommendations.

Rubric area	Evidence in this presentation
Summary and introduction	Executive findings; target definition; project motivation
Collection and wrangling	API, scraping, cleaning and target-mapping slides
EDA visualization and SQL	Scatter, site/orbit bars, annual trends; queries and results
Interactive analytics	Folium inputs/results + companion map; supplied Dash screenshots
Classification and conclusion	Four models, CV/test scores, confusion matrices, model choice
Reproducibility	Repository URL on methodology/result slides and in code package

Remaining evidence gap: add actual Folium basemap screenshots to the geographic slides.

<https://github.com/Cristian210600/SpaceX-Falcon9-Landing-prediction>

Sources and evidence trail

Every quantitative result is tied to a course snapshot or saved notebook output.

Primary project evidence

Seven uploaded notebooks and two dashboard screenshots.

SQL and descriptive rates were recalculated; ML results retain the saved run.

[IBM course datasets](#)

[scikit-learn: common pitfalls and data leakage](#)

[Cover photo: Michael Seeley / NASA APOD](#)